

The rest of this paper is organized as follows. The next section is devoted to the proofs relating minimum degree with the rainbow connection number. In particular, Theorems 1.1, 1.3, and 1.4 are proved. Section 3 consists of the proofs of Theorems 1.5 and 1.6. The final section contains some concluding remarks and open problems, mainly concerning the computational complexity aspects of rainbow connection.

2 Bounded degree graphs

2.1 Minimum degree 3

In this subsection we prove Theorem 1.1. We start with a simple lemma that will be useful in many of the results of this paper.

Lemma 2.1 *If G is a connected graph and $H_1, \dots, H_k$ is a partition of the vertex set of G into connected subgraphs then $rc(G) \leq k - 1 + \sum_{i=1}^k rc(H_i)$.*

Proof: Contracting each H_i to a single vertex we obtain a connected minor of G with k vertices. This minor has rainbow connection number at most $k - 1$. We color each edge connecting vertices in distinct H_i with the color of the corresponding minor edge. For each $i = 1, \dots, k$, the edges inside each H_i are colored with a dedicated set of $rc(H_i)$ colors. ■

Notice that in Lemma 2.1 we allow the H_i to be singletons (and the rainbow connection number of singletons is 0).

The following proposition is an important ingredient in the proof of Theorem 1.1.

Proposition 2.2 *If G is a 2-connected graph with n vertices then $rc(G) \leq 2n/3$.*

Proof: If G is a 5-cycle then $rc(G) = 3$ so the proposition clearly holds in this case. Otherwise, let H be a maximal connected subgraph of G having the property that $rc(H) \leq 2h/3 - 2/3$, where h is the number of vertices of H . We first claim that H exists. Indeed, if G has a triangle then already taking H to be a triangle we obtain $rc(H) = 1 \leq 2 - 2/3$. Otherwise, if G has any cycle of length $k \geq 4$ and $k \neq 5$, then already taking H to be such a cycle we obtain $rc(H) = \lceil k/2 \rceil \leq 2k/3 - 2/3$. Otherwise, if each cycle of G is a C_5 then taking H to be a C_5 attached to one additional edge we obtain $rc(H) = 3$, $h = 6$, and $3 \leq 4 - 2/3$.

We next claim that $h \geq n - 2$. Indeed, assume first that there are three distinct vertices outside of H , say x_1, x_2, x_3 , each having two neighbors in H (the neighbors of x_i do not have to be distinct from the neighbors of x_j). We can add x_1, x_2, x_3 to H and form a larger subgraph H' with $h + 3$ vertices. Suppose e_i, f_i are two edges connecting x_i with H . We use only two new colors to color the 6 designated edges; e_1, e_2, e_3 all get the same color and f_1, f_2, f_3 all get the same color. We now have

$$rc(H') \leq rc(H) + 2 \leq 2h/3 - 2/3 + 2 = 2(h + 3)/3 - 2/3$$

contradicting the maximality of H . It follows that if there are three vertices outside of H then at least one of these vertices, say x , has the property that a shortest path from H to H passing through x has length at least 3 (notice that there must be such a path as the graph is 2-connected). Let, therefore, $a, x_1, \dots, x_t, b$ be a path with $a, b \in H$, with $x_1, \dots, x_t \notin H$, and $t \geq 2$. We can add $x_1, \dots, x_t$ to H and form a larger subgraph H' with $h+t$ vertices. If t is odd we can color the $t+1$ edges of the path with $(t+1)/2$ new colors. In the first half of the path the colors are all distinct, and the same ordering of colors is repeated in the second half of the path. It is straightforward to verify that H' is rainbow connected. If t is even, we can color the $t+1$ edges of the path with $t/2$ colors as follows. The middle edge $(x_{t/2}, x_{t/2+1})$ receives any color that already appears in H . The first $t/2$ edges of the path all receive distinct new colors and in the last $t/2$ edges of the path this coloring is repeated in the same order. Again, it is straightforward to verify that H' is rainbow connected. We now have

$$rc(H') \leq rc(H) + \lceil t/2 \rceil \leq 2h/3 - 2/3 + \lceil t/2 \rceil \leq 2(h+t)/3 - 2/3$$

contradicting the maximality of H .

Having proved that $h \geq n-2$ we now clearly have $rc(G) \leq 2(n-2)/3 - 2/3 + 2 = 2n/3$ as claimed. ■

Although the following asymptotic improvement of Proposition 2.2 does not yield a better bound for Theorem 1.1 it does yield the best possible coefficient of n if we look for an upper bound of the form $cn + o(n)$ on $rc(G)$ for 2-connected graphs.

Theorem 2.3 *If G is a 2-connected graph on n vertices, then $rc(G) \leq n/2 + O(\sqrt{n})$.*

Proof: A suitable edge coloring of G will be constructed sequentially, starting with a cycle on some number n_0 of vertices and making it rainbow connected with $\lceil n_0/2 \rceil \leq n_0/2 + 1/2$ colors. We assume that until the current stage of the procedure, a subgraph G' of G has been made rainbow connected, with n' vertices and $n'/2 + c$ colors.

We have seen in the proof of Proposition 2.2 that if a path P of length t has both endpoints in G' but is internally disjoint from G' , then the t edges of P require as few as $\lceil t/2 \rceil$ new colors to make $G' \cup P$ rainbow connected. This step keeps the value of c unchanged if t is odd, and increases c with $1/2$ if t is even. Our goal is to prove that $c = O(\sqrt{n})$ holds when the entire G is edge-colored.

The key idea is to proceed with adding paths in *non-increasing* order of length. Let t denote the maximum path length that can currently be added, and let $P = a, x_1, \dots, x_{t-1}, b$ be such a path, having its endpoints a, b in G' and its internal vertices x_i outside G' . Observe that no path of length t (or more) and being internally disjoint from $G' \cup P$ can join any x_i to any vertex of $G' \cup P$, otherwise t would not be maximum. Consequently, inserted paths of the same length are internally vertex-disjoint, and once we finish with t , we can never return to length t or more.

If t is odd, the value of c remains the same after path insertion. Suppose that t is even. If just one or two paths of length t can be added before we continue the procedure

with length $t - 1$, then, applying the coloring pattern described earlier, the value of c increases with $1/2$ or 1 , respectively. If three or more paths of length t are added, we multicolor all those paths with the same t new colors (in the same order). In this way a rainbow-connected subgraph is obtained. For three or more paths we insert at least $3t - 3$ internal vertices, but use just $t < (3t - 3)/2$ colors if $t \geq 4$, what makes c decrease. If $t = 2$, c does not increase unless three paths are inserted, in which case c increases with $1/2$.

The worst case for counting this upper bound on $rc(G) - n/2$ is when each even path length $2, 4, \dots, \ell$ occurs precisely two times (and no odd lengths). Even then, $rc(G) - n/2 \leq \ell/2$ holds. Since $\sum_{i=1}^{\ell/2} (2i - 1) < n/2$ is valid for this particular sequence, we obtain $\ell = O(\sqrt{n})$ and the theorem follows. ■

Recall that a graph is *bridgeless* if the removal of an edge does not increase the number of connected components.

Proposition 2.4 *If G is a connected bridgeless graph with n vertices then $rc(G) \leq 4n/5 - 1$.*

Proof: Consider first the case where G is 2-connected. By Proposition 2.2, $rc(G) \leq \lfloor 2n/3 \rfloor \leq 4n/5 - 1$ for all $n \geq 7$. Since a 2-connected graph contains a cycle we have by Lemma 2.1 that $rc(G) \leq n - 2$ and notice that $n - 2 \leq 4n/5 - 1$ for $n = 3, 4, 5$. The only remaining case is, therefore $n = 6$, and we need to show that $rc(G) \leq 3$ in this case. If the longest cycle in G has length 6 then $rc(G) \leq 3$. If the longest cycle has length 5 then G is the graph consisting of the 5-cycle and another vertex of degree 2 whose neighbors are two non-adjacent vertices of the cycle. It is straightforward to check that this graph has $rc(G) = 3$. If the longest cycle has length 4 then G is $K_{2,4}$ which has $rc(G) = 2$. A 2-connected graph with 6 vertices always has a cycle of length at least 4.

Having proved the proposition for 2-connected graphs, we prove it for connected bridgeless graphs by induction on the number of 2-connected components. Let X be the set of vertices of a 2-connected component of G so that X contains only one cut-vertex, say x (recall that such a 2-connected component always exists). Consider the subgraph H of G induced by $(V(G) \setminus X) \cup \{x\}$. It has $n - |X| + 1$ vertices and is connected, bridgeless, and with one less 2-connected component. By the induction hypothesis, $rc(H) \leq 4(n - |X| + 1)/5 - 1$. Since X induces a 2-connected graph, we have $rc(X) \leq 4|X|/5 - 1$. Hence, $rc(G) \leq 4n/5 - 2 + 4/5 < 4n/5 - 1$, completing the proof. ■

Proof of Theorem 1.1: We are given a connected graph $G = (V, E)$ with n vertices and minimum degree at least 3. Let $B \subset E$ denote the set of bridges of G . If $B = \emptyset$ we are done by Proposition 2.4. Thus, we may assume $B \neq \emptyset$. Let $\mathcal{C}$ denote the set of connected components of $G' = (V, E \setminus B)$. Notice that $\mathcal{C}$ contains at least two elements. There are two types of elements in $\mathcal{C}$, singletons and connected bridgeless subgraphs of G . Let $\mathcal{S} \subset \mathcal{C}$ denote the singletons and let $\mathcal{D} = \mathcal{C} \setminus \mathcal{S}$. Each element of $\mathcal{S}$ is, therefore, a vertex, and each element of $\mathcal{D}$ is a subset of vertices that induces a connected bridgeless subgraph.